

Milos MRT 1 Tower

Assembly/Breakdown Instructions



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MILOS MRT 1 Tower is designed for use of original Milos components only

1. Basic specifications, Application(s)

Milos MRT1 – is tower for temporary hanging of Audio/Lighting equipment.

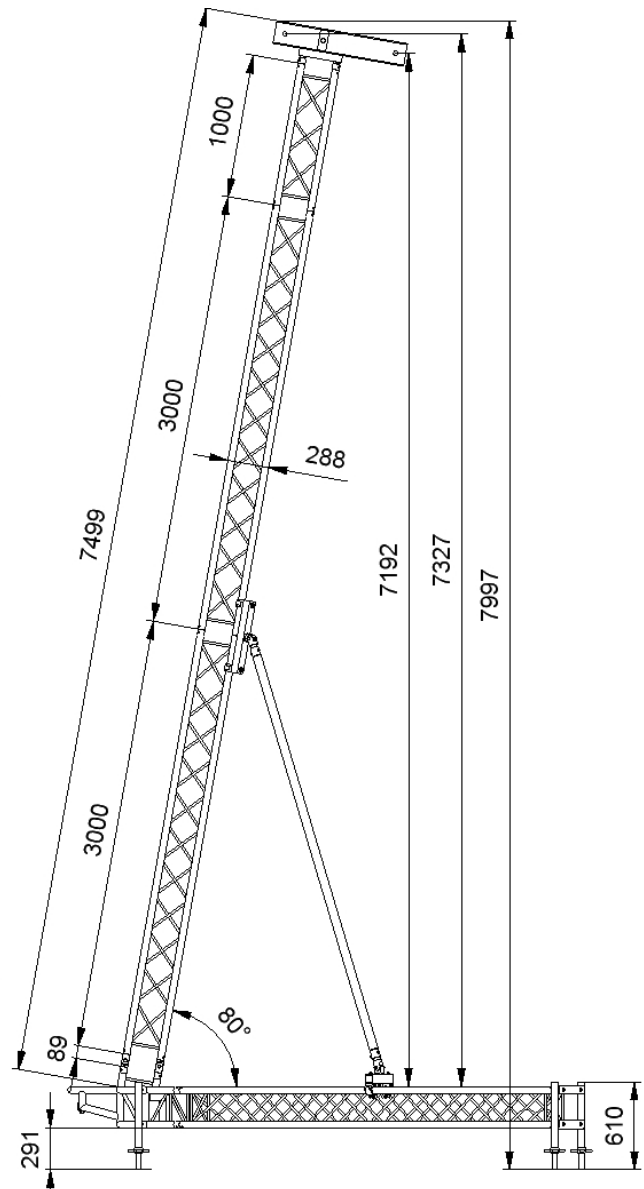
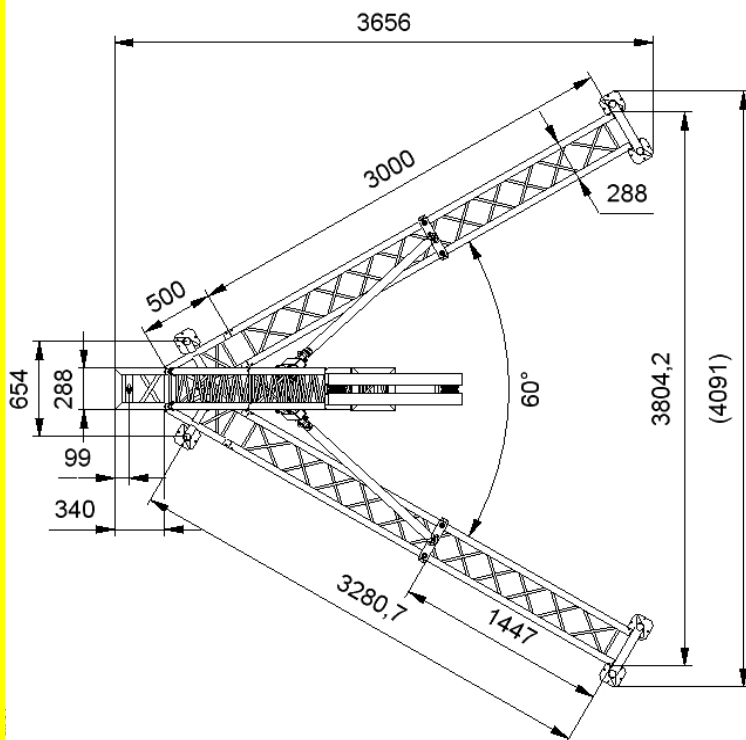
MRT1 incorporates our M290V truss system. The system is an icon of strength and stability, and can be an integral part of your show. Every rigging tower is constructed using our Quick Truss conical connectors for an extremely safe, simple, and secure assembly. The size and weight of the construction makes this ideal for large size events. MRT1 tower can be supplied with various accessories for hoists attachment and ballasts system integration.

Read all instructions before beginning assembly

2. Technical data, Loading charts, Working conditions

Technical data:

Total height:	8 m
Total width:	4 x 3,7 m
Total weight:	180 kg
Capacity:	560 kg



3. MRT 1 tower - components

Milos MRT 1 tower is build from following parts:

Base	Using M290QTV Truss (main tube 48 x 3mm; brace 16 x 2mm), stabiliser attachment horizontal and vertical, QCV corner,
Support for lifting	Stabiliser tube (60 x 6mm), M290QTV Truss (main tube 48 x 3mm; brace 16 x 2mm),
Tower mast	Headsection, M290QTV Truss (main tube 48 x 3mm; brace 16 x 2mm),

MRT 1 Tower

Part description	QTY	Weight	Weight/Total	Drawing position	
MRT1 QTV corner	1	16,145 kg	16,145 kg	1	
MT1 07 hinge part	4	0,631 kg	2,524 kg	2	
QTV 3000	4	18,182 kg	72,728 kg	3	
QTV 1000	1	6,874 kg	6,874 kg	4	
MRT1 headsection	1	19,984 kg	19,984 kg	5	
Screwjack adapters MRT1	2	3,988 kg	7,976 kg	6	
MRT1 stabiliser attachment vertical	1	7,061 kg	7,061 kg	7	
MRT1 stabiliser attachment horizontal - R	1	2,626 kg	2,626 kg	8	
MRT1 stabiliser attachment horizontal - L	1	2,626 kg	2,626 kg	9	
MRT1 stabiliser tube 60 x 6	2	10,772 kg	21,544 kg	10	
Screwjack 40 cm	6	1,532 kg	9,192 kg	11	

4. Important points before installation

Before installation, please make sure that you understand this manual completely.

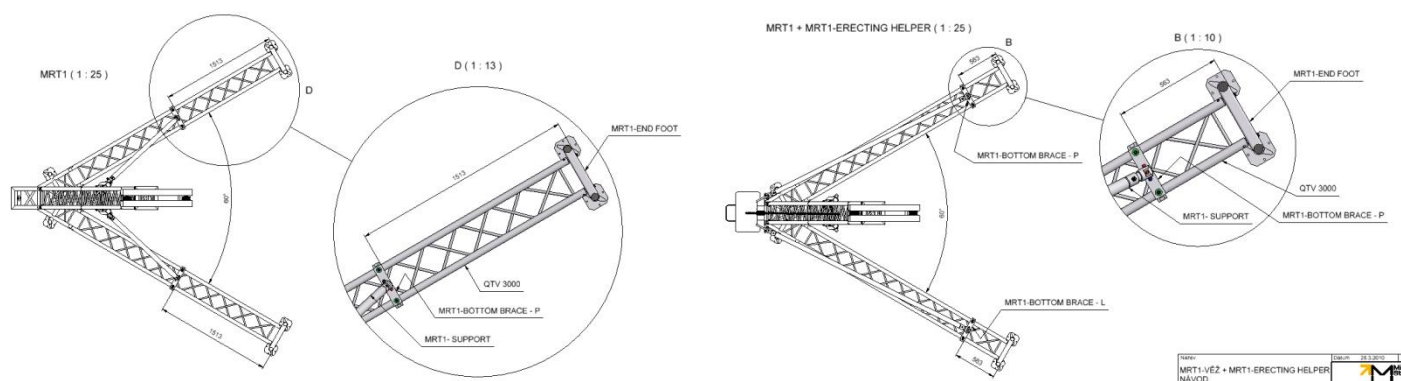
- a)** Specify one person, which will coordinate assembling & breakdown process on site.
- b)** Make sure that you have all parts for assembling before start. In case that some is missing, please contact Milos for spare parts.
- c)** Make sure that the site where the MRT 1 is assembled is stable and can support required loads.
Always use wooden/rubber pads to underlay structures feet.
- d)** Check there are no obstructions within or around the area of installation (including overhead wires, street, trees etc.).
Ensure there is enough space around tower(s) for safe operation with outriggers (short / long) and tower raise.
- f)** All persons, which will work in height over 2,0 m must be trained and wear fall protection equipment.

Prior to assembly check all parts are complete and no items are missing / damaged. Replacement items must be from original of manufacture, use of alternative components is at users own risk.

5. Assembly instruction of Milos tower MRT 1

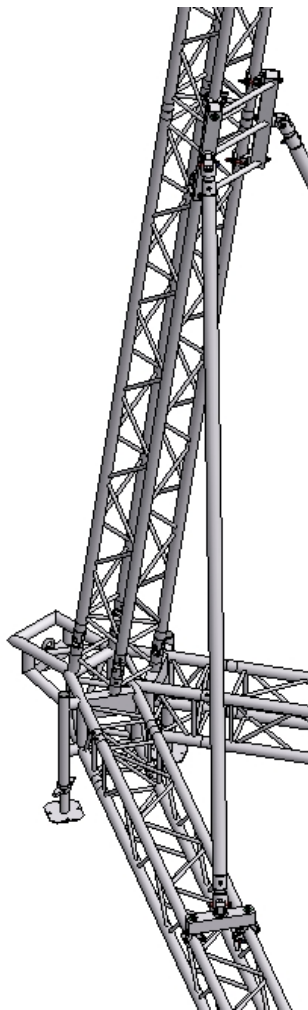
1) Prepare a MRT 1 towers base

- a)** Attach QTV 3000 to conner QCV and properly secure with pin. Check the steel base is horizontal using a spirit level, use screwjacks to adjust and balance as required. Horizontal position of steel base is very important for MRT 1 stability.
- b)** Preparing vertical QTV parts, column section, headsection and stabiliser attachments.

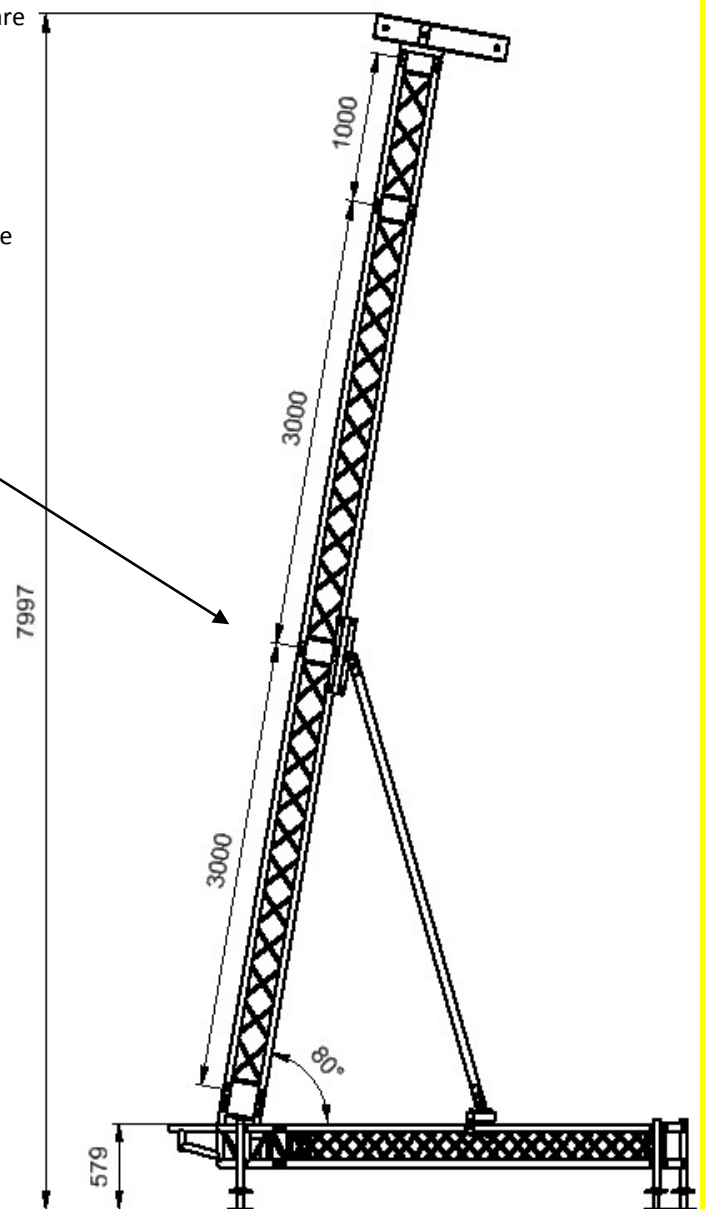


2) Tower column assembling and raising

- a) **Prepare tower mast:** All components QTV connect and prepare stabilise attachments and support tubes.
- b) **Connect tower mast to hinges on base corner.**
The truss tower is now ready for raise. It must be performed with a minimum of 2 people.
- c) **Support to tower:** Lift the tower mast and secure pins on base corner. After that use support tube for securing tower mast. Place for support tube is on tower at the join.

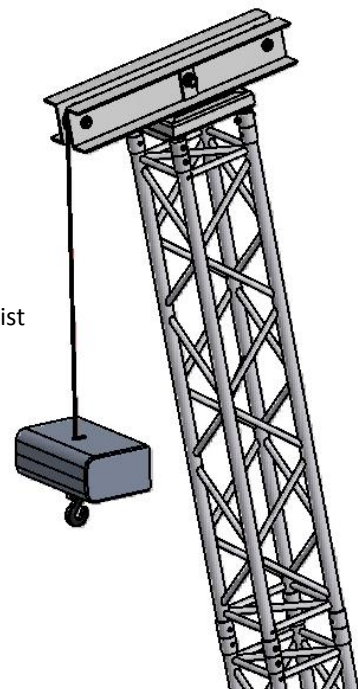


- d) Next step is installing lead chain/manual hoist to the headsection.



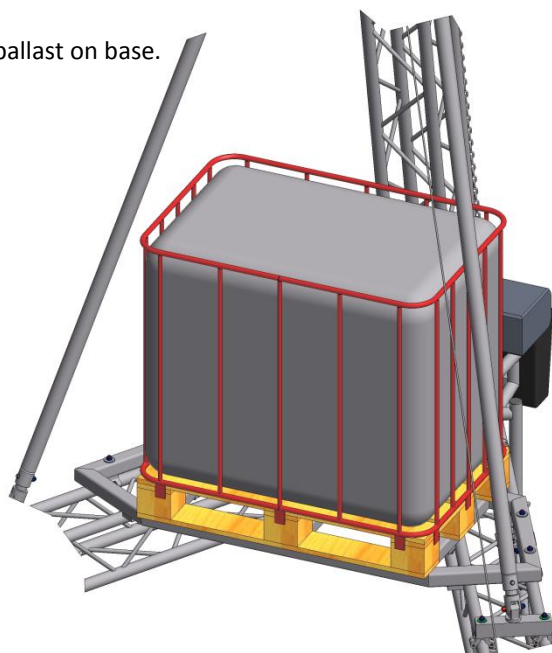
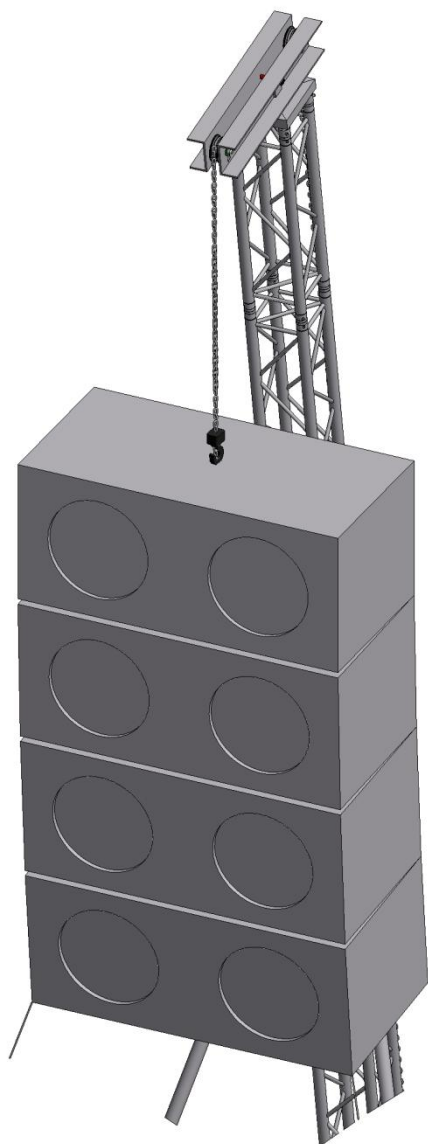
3) Securing tower

- a) For maximum loading capacity MRT 1 tower connect lead section/manual hoist to front pin of the Headsection.

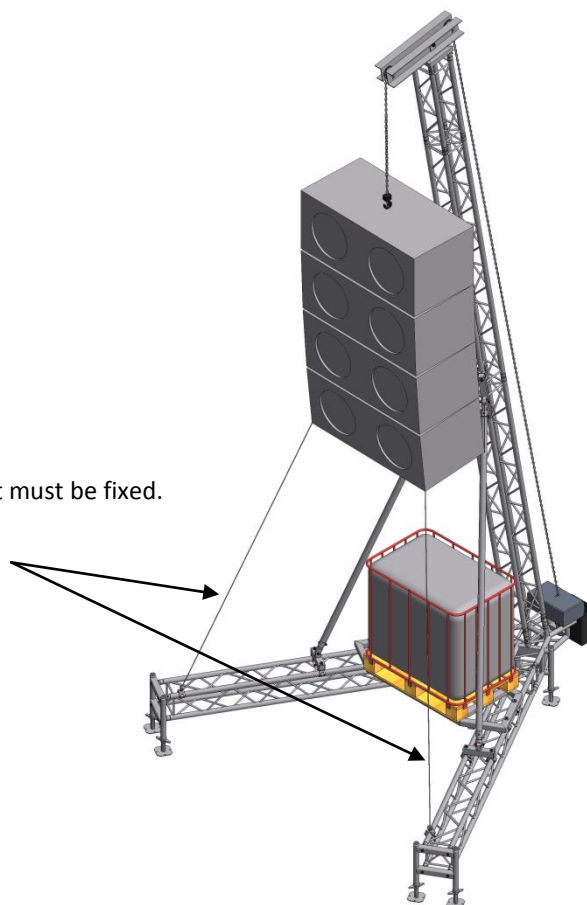


b) Securing the tower add the ballast on base.

c) The Tower is now ready for speaker lifting.



d) All equipment must be fixed.



In keys of need contact Milos or report to statics

7. Working conditions

MRT 1 Tower is certified for max. wind speed 74 km/h according DIN 4112 (temporary buildings).
For loading capacity of the MRT 1 Tower, please refer to Static report.

Operation of tower:

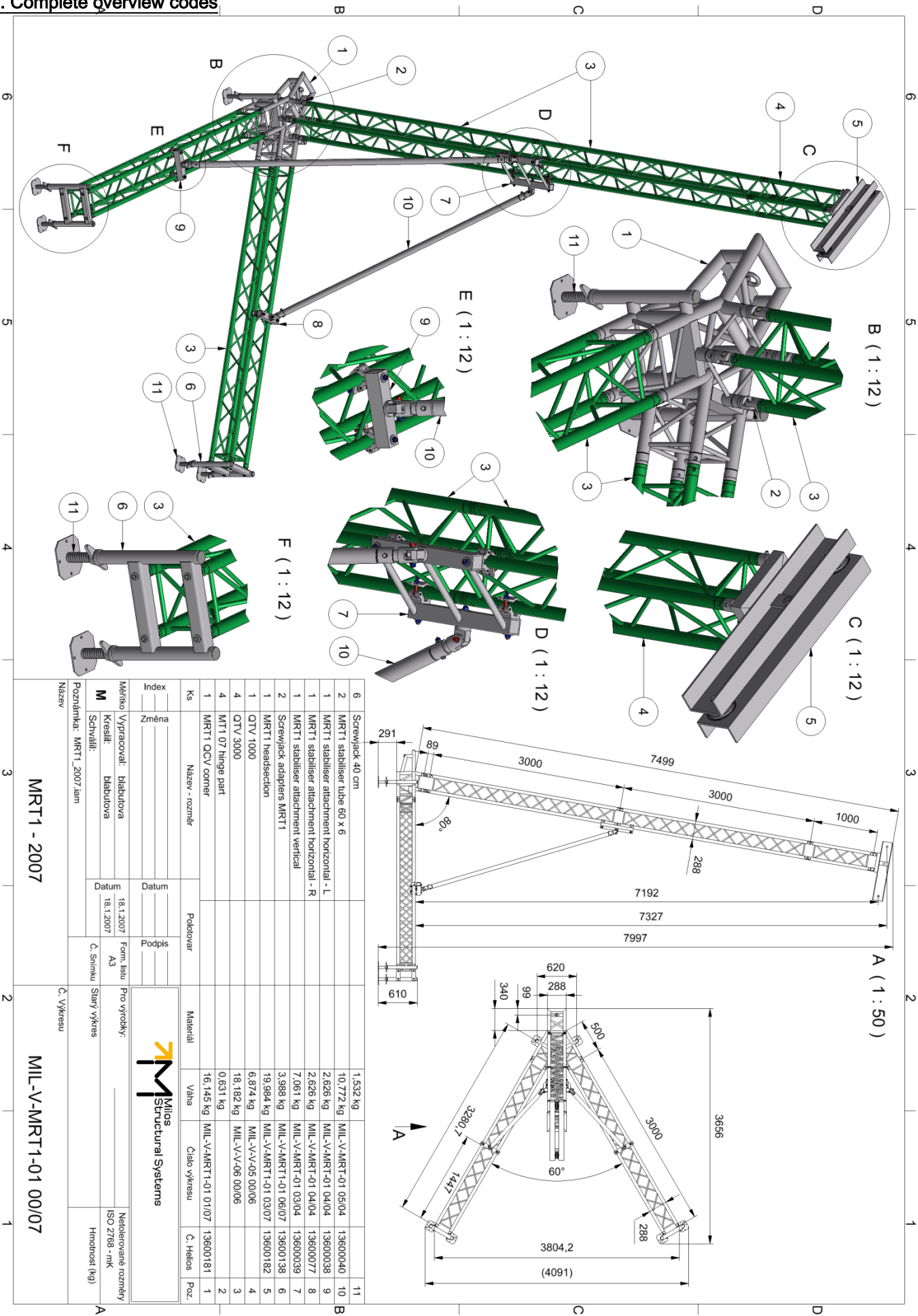
MRT 1 is tower for temporary hanging Audio and Lighting equipments and there must be a trained person, which will be responsible for security during assembling, whole show and dismantling.

**** IMPORTANT NOTICES ****

1. Always use protective equipment and professional tools.
2. When securing sleeve block, or working at any heights, always use professional safety harnesses tools that will protect you in case of fall.
3. Inspect properly every connection and make sure that all bolts are secure, and r-clips and fixed.
4. If using electrical equipment on / with tower(s) ensure tower is properly earthed.
5. Before each assembly, check all items and make sure they are not incorrect or showing any signs of wear or visual damages - NEVER use damaged or worn-out parts.

Users Notes list		

8. Complete overview codes



9. Breakdown instructions for MRT1 Tower

1. Before breakdown check space under tower and make clear from all equipment or obstacles. Make visual check of tower parts and connections for any complications.

2. Release wires: Remove tower ballast, unlock a safety wires on tower base.

3. Column lowering: When chain is used, we are recommending disconnect and carefully remove from column. Release cable to prevent any problems with lowering. Lowering of column must be performed with a minimum of two people.

4. Lowering tower mast: Must disassemble support tube and two secure pins on corner base. Slowly run tower mast to down.

5. Disassemble: Now you can disassemble all components. First tower mast and after others.

Before releasing the two outer hinge parts, check there are no obstacles in the direction the lowering column. Release ONLY the two outer hinge parts (opposite side to lowering side). Once more check area for any unauthorised people, obstructions – once clear slowly lower column to ground. NEVER attempt to lower with less than the recommended no. of persons (3 or more).

Store all connection parts (aluminium connectors, steel pins, r-clips, bolts screws etc) in a safe and dry location. Mechanical parts must be checked periodically and lubricated where necessary. Maintenance to lifting hoist must be followed according to manufacturers recommendations. Never attempt assembly with any damaged or missing components, for spares / replacements please contact one of our sales offices or distributors.

On behalf of Milos Structural Systems, we would like to thank you for safely using our products.

We welcome any feedback or suggestions and look forward to hearing from you.